Frequency-Domain Visual Prompting for Test-Time Adaptation

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Abstract

We propose a frequency-conditioned prompting method for test-time adaptation to improve performance when test images differ from training data distribution.Our method keeps the backbone model frozen and learns only a small prompt from each input image.First, we extract radial frequency features from the Fourier amplitude spectrum.Then, a lightweight MLP uses these features to generate an input-specific prompt, which modifies the amplitude while keeping the phase unchanged.The adapted image is then reconstructed and passed to the frozen model for prediction.We evaluate the method on PACS and CIFAR-10-C using ResNet-18.On CIFAR-10-C, it improves performance on some corruptions, especially shot noise and zoom blur, showing that the approach is effective for image shifts with clear frequency patterns.On PACS, the improvements are small and not consistent, and the method performs worse in the online setting.Overall, the results suggest that frequency-based prompting is a useful and lightweight direction for image recognition under distribution shift, but more work is needed to handle complex changes such as style and semantic variation.The code is availble

1. Introduction

Deep learning models often suffer from performance drops when deployed on data that differs from the training distribution.This problem, known as domain shift, is common in real-world applications.Test-time adaptation (TTA) aims to address this issue by adapting models during inference without access to source data [2].While existing methods such as entropy minimization [3] update model parameters at test time, they can be unstable and computationally expensive.

Recent approaches explore prompt-based methods, where small input-level modifications are used to adapt the model while keeping the backbone fixed. Visual Prompt Tuning (VPT) [1] and its extensions show that learnable prompts can effectively adapt models without modifying internal parameters. More recently, VPTTA [4] applies prompting to test-time adaptation. At the same time, frequency-domain representations provide a useful way to understand how different types of corruptions affect images. In particular, many common corruptions such as noise and blur can be interpreted as changes in the frequency spectrum, where amplitude relates to appearance and phase captures structure.

Despite these advances, existing approaches have some limitations.Methods that update model parameters during test time can be sensitive to noisy predictions and may lead to unstable behavior. On the other hand, prompt-based methods, while efficient, often operate in the input or feature space without explicitly modeling how different types of corruptions affect the image. In particular, many corruptions have a strong frequency-based structure, which is not directly exploited in most existing methods.

This motivates the need for a more structured way to perform adaptation by explicitly using frequency-domain information. By working in the frequency domain, it becomes possible to selectively modify different components of the image, such as low- and high-frequency regions, in a controlled manner. This provides a simple and interpretable way to handle different types of distribution shifts without modifying the backbone model.

Motivated by these observations, we propose a simple frequency-conditioned prompting method for test-time adaptation. Instead of directly modifying model parameters, we extract frequency-based features from the input image and use a lightweight MLP to generate a prompt. This prompt is used to modify the amplitude of the image in the frequency domain, while keeping the phase unchanged, allowing us to adapt the input in a structured way.

We evaluate our method on PACS and CIFAR-10-C under different settings. Our results show that the method is effective for certain types of corruptions, particularly those with clear frequency characteristics such as noise and blur.

Our contributions can be summarized as follows:

Code available at:https://github.com/vhemanth06/FreqPrompt-TTA

- We propose a frequency-conditioned prompting method for test-time adaptation that operates in the frequency domain.
- We introduce a simple MLP-based approach to generate input-dependent prompts using radial frequency features.
- We show that the method is effective for certain corruption types while maintaining performance comparable to the baseline without modifying the backbone model.

2. Related Work

2.1. Test-Time Adaptation

Test-Time Adaptation (TTA) focuses on improving model performance under distribution shifts without access to source data.A common approach is entropy minimization [3], which encourages confident predictions on unlabeled test samples. While effective, such methods often update model parameters directly, which can lead to instability and time overheads during inference.

2.2. Prompt-Based Adaptation

Prompt-based methods provide a lightweight alternative by adapting inputs instead of model parameters. Visual Prompt Tuning (VPT) [1] shows that learnable prompts can adapt models while keeping the backbone fixed.More recently, VPTTA [4] extends this idea to test-time adaptation by generating input-specific prompts. It also highlights the role of frequency information, particularly low-frequency components, in handling domain shifts.

2.3. Frequency-Based Methods

Frequency-domain representations have been widely used to analyze images.In particular, amplitude is related to appearance variations such as noise and blur, while phase captures structural information. Prior work such as VPTTA implicitly uses frequency information for adaptation.

In this project, we build on these ideas and explore a more structured approach by using radial frequency features and an MLP to generate input-dependent prompts. This allows us to adapt different frequency bands in a controlled manner during test-time adaptation.

3. Method

3.1. Overview

We propose a frequency-conditioned prompting method for test-time adaptation (TTA). Given an input image, we extract frequency-based features and use a lightweight MLP to generate an input-specific prompt. This prompt is used to modify the amplitude of the image in the frequency domain, and the adapted image is then passed through a frozen backbone model.

3.2. Radial Frequency Feature Extraction

Given an input image $x \in \mathbb{R}^{C\times H\times W}$, we compute its Fast Fourier transform:

$$\hat{x} = \mathcal{F}(x), \tag{1}$$

followed by a frequency shift to center low-frequency components.

We decompose $\hat{x}$ into amplitude A and phase ϕ:

$$A = |\hat{x}|, \quad \phi = \angle\hat{x}. \tag{2}$$

The amplitude represents the strength of different frequency components and is related to appearance changes such as noise, blur, and texture, while the phase captures structural information like edges and layout. Recent papers like VPTTA [4] has explored frequency-domain prompting by modifying the amplitude while keeping the phase unchanged. Inspired by this, we follow the same idea in our proposed method.

To obtain a compact representation for the MLP, we divide the frequency domain into N radial bins and compute the mean energy in each bin:

$$e_{c,n} = \frac{1}{|\mathcal{R}_n|} \sum_{(u,v)\in\mathcal{R}_n} A_c(u,v), \tag{3}$$

where $\mathcal{R}_n$ denotes the n-th radial region.The bins are defined based on their distance from the center of the frequency spectrum, where the 0-th bin represents low-frequency components, and higher-index bins capture higher-frequency components, with the $(N-1)$-th bin representing the highest frequencies. This results in a feature vector $e \in \mathbb{R}^{C\times N}$.

3.3. Prompt Generation

The radial energy features are flattened and passed through a lightweight MLP:

$$p = \text{MLP}(e), \tag{4}$$

where $p \in \mathbb{R}^{C\times N}$ represents the modulation coefficients for each radial bin and channel.

The prompt is then expanded back to the full frequency domain using predefined radial masks. Each bin-wise coefficient is assigned to all frequency locations belonging to its corresponding radial region:

$$P_c(u,v) = \sum_{n=0}^{N-1} p_{c,n} \cdot M_n(u,v), \tag{5}$$

where $P_c(u,v)$ denotes the prompt at frequency location (u,v) for channel c, and $M_n(u,v)$ is a binary mask that equals 1 if (u,v) lies in the n-th radial bin and 0 otherwise.

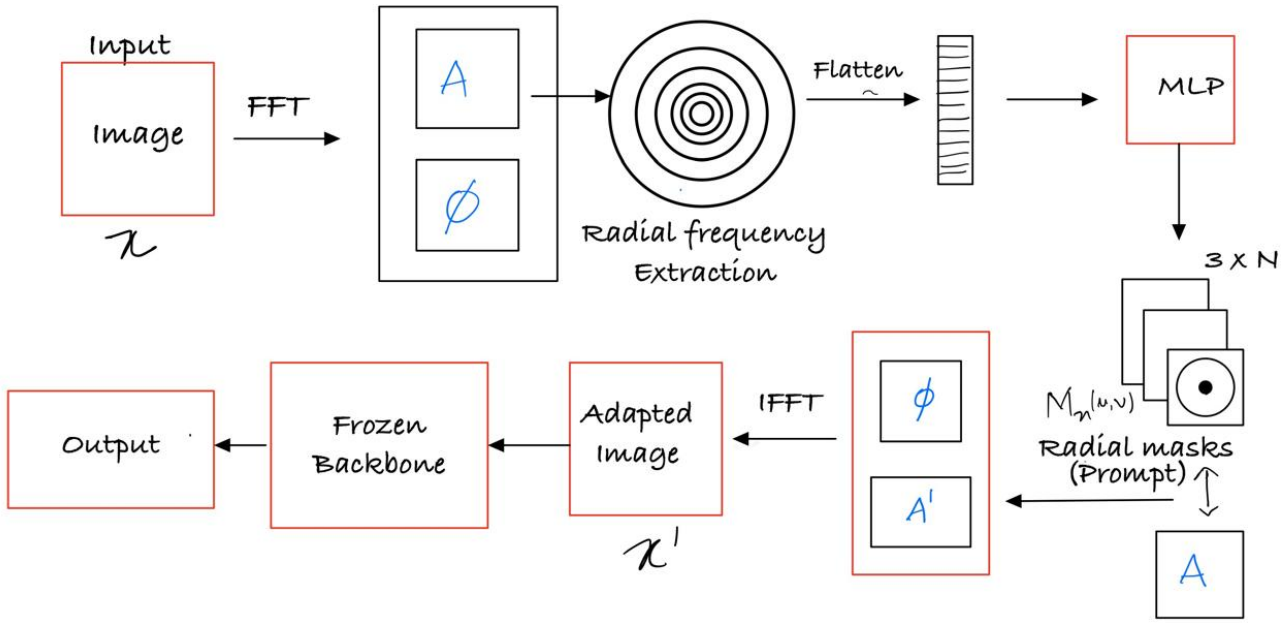

Figure 1. Overview of our method. The input image is converted to the frequency domain using FFT, where amplitude and phase are obtained. Radial frequency features are extracted and passed through an MLP to generate a prompt, which is used to modify the amplitude. The adapted image is then reconstructed using inverse FFT and passed through a frozen backbone model to obtain predictions. The output is used to compute an entropy minimization loss with an ℓ_2 penalty, which is then backpropagated to update the MLP.

3.4. Frequency-Domain Adaptation

We apply multiplicative modulation to the amplitude spectrum:

$$A' = A \cdot (1 + \alpha P), \tag{6}$$

where α is a learnable scaling parameter.

This formulation preserves the original amplitude when the scaling parameter α is close to zero, so the adapted image remains similar to the input. This helps avoid unwanted distortions and keeps the adaptation stable during test-time.

The adapted signal is reconstructed for each channel using the original phase:

$$\hat{x}'_c = A'_c \cdot (\cos \phi_c + i \sin \phi_c), \tag{7}$$

and then converted back to the image domain using the inverse Fast Fourier Transform:

$$x'_c = \mathcal{F}^{-1}(\hat{x}'_c). \tag{8}$$

The reconstructed channels are combined to form the adapted image x', which is then passed through a frozen backbone model.

3.5. Test-Time Adaptation Objective

We optimize the prompt-generating MLP using entropy minimization:

$$\mathcal{L}_{ent} = -\sum_k p_k \log p_k, \tag{9}$$

where p_k are the predicted class probabilities from the frozen backbone model.Entropy minimization encourages confident predictions on unlabeled test samples and has been shown to be effective for test-time adaptation [3].

To control the prompt magnitude, we include an ℓ_2 penalty:

$$\mathcal{L} = \mathcal{L}_{ent} + \lambda \|p\|_2^2. \tag{10}$$

This regularization helps prevent large changes to the input and improves stability during adaptation.

During adaptation,only the MLP parameters and the scaling factor α are updated, while the backbone remains frozen.

4. Experiments

4.1. Datasets

We evaluate our method on two datasets: PACS and CIFAR-10-C.

PACS: PACS is a domain generalization dataset with four domains: Photo, Art Painting, Cartoon, and Sketch. It contains 7 classes, with 1,670 images in Photo, 2,048 in Art Painting, 2,344 in Cartoon, and 3,929 in Sketch. The domains have large visual differences, making it useful for testing performance under domain shifts. Following VPTTA [4], we treat one domain as the target and use the remaining three as source domains for training.

CIFAR-10-C: CIFAR-10-C is a corrupted version of CIFAR-10 used to evaluate robustness under different types of noise and distortions.It contains multiple corruption types at 5 severity levels.In this project, we evaluate on five corruptions: shot noise, fog, snow, brightness, and zoom blur, using severity level 3.

4.2. Implementation Details

We use a ResNet-18 backbone pre-trained on the source domain.For PACS, the backbone is trained on three domains, with the remaining domain treated as the target. During evaluation, the target domain samples are shuffled before adaptation. For CIFAR-10-C, the backbone is trained on the standard CIFAR-10 dataset. The backbone remains frozen during test-time adaptation. Images are resized to 224×224.

For frequency feature extraction, we use $N = 16$ radial bins. The prompt generator is a lightweight MLP with two hidden layers of size 256. The scaling parameter α is initialized to a small value to ensure stable updates.

During test-time adaptation, we optimize only the MLP parameters and α using the AdamW optimizer. The learning rate for the prompt generator is set to 2.5×10^{-2}. We use entropy minimization along with an ℓ_2 penalty on the prompt vector with the regularization constant of a value 0.5. The scaling parameter α is initialized to 0.07. These hyperparameters were selected based on empirical tuning.

4.3. Evaluation Protocol

We perform test-time adaptation in a mini-batch setting, where the model is updated on batches of target domain data without access to source data. A batch size of 32 is used for adaptation. We also evaluate a online setting using a batch size of 1. Performance is measured using classification accuracy.

Method	Art	Cartoon	Photo	Sketch
Base	79.74	75.00	95.93	73.04
Ours	79.83	74.87	95.93	73.30

Table 1. Results on PACS in mini-batch adaptation (batch size = 32)

Method	Art	Cartoon	Photo	Sketch
Base	80.23	75.00	95.93	73.04
VPTTA	81.49	78.41	96.31	76.46
Ours	79.58	73.29	95.93	72.43

Table 2. Results on PACS in online adaptation(batch size = 1)

Corruption	Base (%)	Ours (%)
Shot Noise	50.07	66.43
Snow	85.27	85.35
Fog	93.25	93.41
Brightness	93.65	93.73
Zoom Blur	76.87	80.36

Table 3. Results on CIFAR-10C (severity level 3)

We compare the following settings:

- **Base**: Backbone model without adaptation
- **Ours**: Frequency-conditioned prompting with test-time adaptation
- **VPTTA**: Prompt-based test-time adaptation baseline

4.4. Results on PACS

Mini-batch setting (Batch size = 32):

The results are shown in Table 1. In this setting, our method shows only small improvements in the Art and Sketch domains, while performance slightly decreases on Cartoon. For the Photo domain, the baseline performance is already high, and the adapted outputs remain similar, resulting in no significant change. The decrease in the Cartoon domain suggests that the method may not fully capture the domain shift in this case.

Online setting (Batch size = 1):

The results are shown in Table 2. In this setting, our method does not show improvement over the baseline and in some cases leads to a slight decrease in performance. For the Photo domain, there is no significant change in accuracy, as the baseline model already performs well.

4.5. Results on CIFAR-10-C

We further evaluate our method on CIFAR-10-C across multiple corruption types. As shown in Table 3, our method improves performance over the baseline, especially for corruptions like shot noise and zoom blur. For example, accuracy on shot noise increases from 50.07% to 66.43%, and on zoom blur from 76.87% to 80.36%.

For other corruptions such as fog and brightness, the improvements are small.This suggests that these types of shifts are not fully captured by the current frequency-based approach. Overall, the improvements are more noticeable for corruptions with strong frequency characteristics, while global changes such as brightness and fog show limited gains.

Overall, the method performs better for corruptions that have clear frequency patterns, while it is less effective for global or semantic shifts.

4.6. Qualitative Results

We show qualitative results under shot noise corruption in Fig. 2. The adapted images show reduced noise compared to the original inputs.

We also visualize an example of the learned frequency-domain prompt in Fig. 3.

More qualitative examples can be found in the project Github repository.

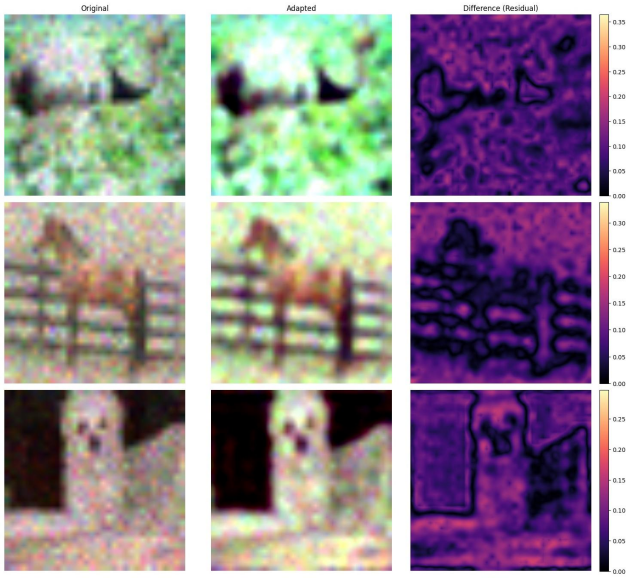

Figure 2. Qualitative results under shot noise corruption. From left to right: original input, adapted image, and the difference. The method mainly modifies high-frequency components corresponding to noise.

5. Limitations

While our method is simple and structured, it has some limitations.

First, the improvements are not consistent across datasets and domains.While we see gains on some corruption types in CIFAR-10-C, the method shows little or no improvement on PACS and sometimes even a slight drop in performance.

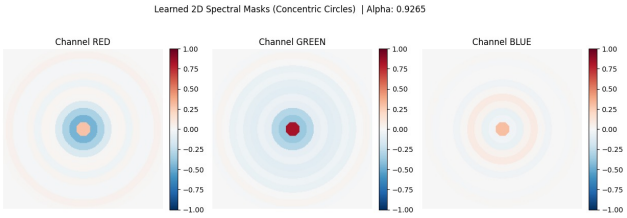

Figure 3. Example of learned prompt under shot noise corruption

Second,the method mainly relies on frequency-based changes, which may not capture more complex domain shifts.In particular, global or semantic differences, such as style variations in PACS, are not handled well.

Third,the method performs worse in the online setting, suggesting that it works better in batch-based adaptation.

Finally, our experiments are limited to a single backbone (ResNet-18) due to compute and time constraints.Testing on larger models and different architectures could give a better understanding of the method.

6. Conclusion

In this work,we proposed a frequency-conditioned prompting method for test-time adaptation. Our approach uses radial frequency features and a lightweight MLP to modify the amplitude while keeping the phase unchanged.

We evaluate the method on PACS and CIFAR-10-C. The results show that it works well for certain corruptions like noise and blur, while maintaining stable performance overall in CIFAR-10C.

However,the improvements are not consistent across all domains(like PACS), indicating that frequency-based adaptation alone may not be sufficient for more complex shifts.

Future work would include exploring better frequency features,improving hyperparameter tuning, and testing the method on other models and datasets.

References

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